IT Infrastructure Security Audit – Rajam Medical Centre & Hospital

Prepared For : Rajam Medical Centre & Hospital

Prepared By : Sharumathi S B

Audit Type : Cyber Security Infrastructure Audit

Audit Period : March 2026

Location : Rajam Medical Centre & Hospital, Coimbatore

Certificate / Declaration

This report presents the results of the Cyber Security Audit conducted for Rajam Medical Centre & Hospital. The audit was carried out to assess the current security posture of the hospital's IT infrastructure, identify vulnerabilities, and recommend improvements to strengthen the cybersecurity framework.

All findings presented in this report are based on system analysis, infrastructure review, and cybersecurity best practices.

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Date:
March 2026

Acknowledgement

I would like to express my sincere gratitude to the management and staff of Rajam Medical Centre & Hospital for providing the necessary support and access to conduct this Cyber Security Audit.

Special thanks to the hospital administration for allowing the evaluation of their IT infrastructure and supporting the cybersecurity awareness initiative.

Executive Summary

This report presents the results of the Cyber Security Audit conducted for Rajam Medical Centre & Hospital. The purpose of this audit was to analyze the hospital's IT infrastructure, evaluate its cybersecurity posture, and identify potential security risks.

The audit examined various components of the hospital's digital infrastructure including network security, system security, IoT medical devices, access control, and incident management procedures.

The audit findings indicate that while the hospital maintains basic security measures, several areas require improvement such as patch management, network segmentation, and user access control.

The overall cybersecurity compliance level was assessed at approximately 60%, indicating a moderate security posture that requires further improvements.

Objective of the Audit

The main objectives of this cyber security audit were:

• To assess the current security posture of the hospital's IT infrastructure.

• To identify vulnerabilities and security risks in the existing systems.

• To evaluate the effectiveness of current security controls.

• To ensure protection of sensitive patient data and hospital systems.

• To provide recommendations for improving cybersecurity practices.

Audit Methodology

The cyber security audit was conducted using a structured methodology that involved multiple stages of evaluation and analysis. Initially, the hospital’s IT infrastructure was reviewed to understand the architecture of systems and networks used within the organization. Security configurations of critical systems were analyzed to determine their level of protection against potential threats. The patch status of systems and medical devices was reviewed to verify whether software updates and security patches were properly implemented. Network segmentation and firewall configurations were examined to evaluate the effectiveness of network protection mechanisms. User access privileges were also reviewed to determine whether proper authentication and authorization controls were in place. Finally, incident records and security logs were analyzed to understand past security events and evaluate the hospital’s incident response capability.

Data Collection and Audit Dataset

During the cyber security audit, relevant data was collected from system reviews, network analysis, and infrastructure assessments within the hospital. The dataset includes system security status, network segmentation information, medical IoT device statistics, user access control details, and recorded security incidents. This dataset was used to evaluate the overall security posture of the hospital and identify potential vulnerabilities in the IT infrastructure.

1. Audit Overview Dataset

This dataset summarizes the overall audit statistics collected during the cyber security assessment.

Metric	Value
Total Systems Audited	7
Total Security Controls Reviewed	48
Compliance Rate	60%
Critical Vulnerabilities	7
High Risk Issues	9
Medium Risk Issues	12
Low Risk Issues	20
Audit Period	March 2026

2. System Security Dataset

This dataset contains the security status of the hospital's major IT systems.

System Name	Security Status	Patch Status	Risk Level
Electronic Medical Record (EMR)	Needs Improvement	Outdated	High
Laboratory Information System	Secure	Updated	Low
Firewall System	Secure	Updated	Low
Security Monitoring System	Partial	Pending	Medium

IoT Medical Devices	Vulnerable	Outdated	High
Nurse Station Computers	Secure	Updated	Low
Mobile Access Devices	Partial	Pending	Medium

3. Network Security Dataset

This dataset shows different network segments and their risk levels.

Network Segment	VLAN ID	Number of Devices	Risk Score
Administration Network	10	28	Low
EMR System Network	20	45	High
Guest WiFi Network	30	120	Critical
Laboratory Devices Network	40	32	Medium
Medical IoT Devices Network	50	67	High

4. IoT Medical Device Dataset

This dataset contains information about medical equipment connected to the network.

Device Type	Category	Number of Devices	Firmware Status	Risk Level
Patient Monitors	Monitoring	18	Outdated	High
Digital X-Ray Systems	Imaging	4	Outdated	High
Infusion Pumps	Therapy	24	Pending	Medium
Laboratory Analyzers	Diagnostic	8	Updated	Low
Ultrasound Machines	Imaging	6	Pending	Medium
ECG Machines	Cardiology	7	Updated	Low

5. Access Control Dataset

This dataset shows user account and access control statistics.

Category	Count
Total User Accounts	312

Active Users	270
Inactive Users	42
Admin Privilege Accounts	18
MFA Enabled Accounts	0

6. Security Incident Dataset

This dataset records cyber security incidents identified during the audit period.

Incident ID	Incident Type	Severity	Response Time
INC001	Malware Attempt	Critical	1.5 Hours
INC002	Unauthorized Access Attempt	High	3 Hours
INC003	Suspicious Network Activity	Medium	2 Hours
INC004	Phishing Email Report	Medium	4 Hours
INC005	System Login Failure	Low	1 Hour

Systems and Infrastructure Reviewed

During the audit, several critical systems and infrastructure components were reviewed. These included the Electronic Medical Record system, which stores and manages patient health information. The Laboratory Information System was examined to evaluate the security of diagnostic data and lab operations. Network infrastructure components such as firewalls and internal network segments were also reviewed. In addition, nurse station computers, mobile access devices, and other workstation systems were assessed for security compliance. Medical Internet of Things devices such as patient monitors, infusion pumps, imaging equipment, and diagnostic machines were also reviewed as they form a significant part of the hospital's digital environment.

System Security Checklist

Audit Item	**Status**	**Remarks**
Operating System Updates	Partial	Some systems not patched
Antivirus Protection	Implemented	Active on workstations

Audit Item	Status	Remarks
Firewall Configuration	Implemented	Properly configured
Security Monitoring	Partial	Limited monitoring
Backup Systems	Partial	Verification needed

Network Security Assessment

The hospital's network infrastructure was analyzed to determine whether proper security measures were implemented to protect internal systems. The network consists of different segments used to manage administrative systems, clinical systems, and guest network access. The audit identified that although network segmentation exists, some medical devices and internal systems are connected within shared network environments. This configuration may increase the risk of unauthorized access or malware propagation across systems. The guest WiFi network also requires stronger security controls to ensure that external users cannot access internal hospital systems.

Network Security Checklist

Audit Item	Status	Remarks
Network Segmentation	Partial	IoT devices not isolated
Guest WiFi Security	Needs Improvement	Additional controls required
Router Security	Implemented	Basic protections active
VPN Access	Not Implemented	Recommended for remote access

Medical IoT Device Security

Medical IoT devices are widely used in modern healthcare facilities to monitor patients and support clinical operations. These devices include patient monitoring systems, imaging equipment, infusion pumps, and diagnostic machines. During the audit, several IoT devices were found to be operating with outdated firmware versions. Outdated firmware can expose devices to cybersecurity vulnerabilities that attackers may exploit. It is important

for the hospital to regularly update device firmware and ensure that these devices are connected to secure network segments to minimize potential risks.

Access Control Evaluation

Access control mechanisms were reviewed to assess how user accounts and permissions are managed within the hospital's systems. The audit found that multiple user accounts remained active even though they were no longer in use. In addition, administrative privileges were assigned to several users, which may increase the risk of unauthorized modifications to critical systems. Another important observation was the absence of multi-factor authentication for administrative access. Implementing stronger authentication mechanisms would significantly enhance system security and reduce the risk of unauthorized access.

Access Control Checklist

Audit Item	Status	Remarks
Password Policy	Weak	Needs stronger policy
User Access Review	Partial	Inactive accounts present
Multi Factor Authentication	Not Implemented	Recommended
Administrator Privileges	Moderate Risk	Too many admin accounts

Incident Management Review

The hospital maintains records of cybersecurity incidents and system alerts. During the audit, a few incidents such as malware attempts and unauthorized access attempts were identified. Although the hospital has basic procedures to respond to security incidents, improvements are required in monitoring capabilities and response time. Implementing advanced monitoring systems and automated alerts would help the hospital detect and respond to cyber threats more effectively.

Key Findings

The cyber security audit identified several important findings that require attention. Some critical systems were found to be running outdated software versions without the

latest security patches. Medical IoT devices were not completely isolated within secure network segments, increasing potential exposure to network threats. Backup verification processes were not conducted frequently, which may create challenges during data recovery situations. Additionally, password policies were not strong enough to prevent unauthorized access, and multi-factor authentication was not implemented for sensitive systems.

Risk Assessment

Based on the audit findings, the overall cybersecurity risk level for the hospital was categorized as moderate. While the hospital has implemented some basic security controls, several areas require improvement to ensure better protection of patient data and hospital systems. Key risk areas include patch management, network segmentation, access control policies, and security monitoring. Addressing these risks will significantly improve the hospital's cybersecurity resilience.

This section is very important in every cyber security audit report.

Risk ID	Risk Description	Impact	Risk Level	Recommendation
R1	Unpatched EMR system	Data breach risk	High	Apply latest security patches
R2	Outdated IoT device firmware	Device vulnerability	High	Update firmware regularly
R3	Weak password policy	Unauthorized access	Medium	Enforce strong password rules
R4	No multi-factor authentication	Account compromise	High	Implement MFA
R5	Inactive user accounts	Unauthorized access	Medium	Disable inactive accounts

Security Maturity Score

Security Area	Score
Network Security	65%

Security Area	Score
System Security	60%
Access Control	55%
Incident Management	58%
Overall Security Score	**60%**

Recommendations

Several recommendations were developed to improve the cybersecurity posture of the hospital. Immediate actions include disabling inactive user accounts, verifying backup integrity, and applying security patches to critical systems. The hospital should also strengthen password policies and implement multi-factor authentication for administrative access. In the short term, it is recommended that the hospital improve network segmentation for medical devices and conduct regular vulnerability assessments. In the long term, the hospital should consider implementing centralized security monitoring systems and conducting regular cybersecurity awareness training programs for staff members.

Conclusion

The cyber security audit conducted at Rajam Medical Centre & Hospital provided valuable insights into the current security posture of the hospital's IT infrastructure. While several security measures are already in place, there are areas that require improvement to enhance overall protection. By implementing the recommendations outlined in this report, the hospital can significantly strengthen its cybersecurity framework and ensure better protection of patient data, medical systems, and digital operations. Continuous monitoring, regular system updates, and cybersecurity awareness among staff will play a crucial role in maintaining a secure healthcare environment.